Passive π Filter Analysis

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Analysis

Select a value for L in the π filter showed in figure 1 so that $V_o = \frac{1}{10}V_{in}$ at $f = 100\,\text{Hz}$. Select $C = 10\,\mu\text{F}$

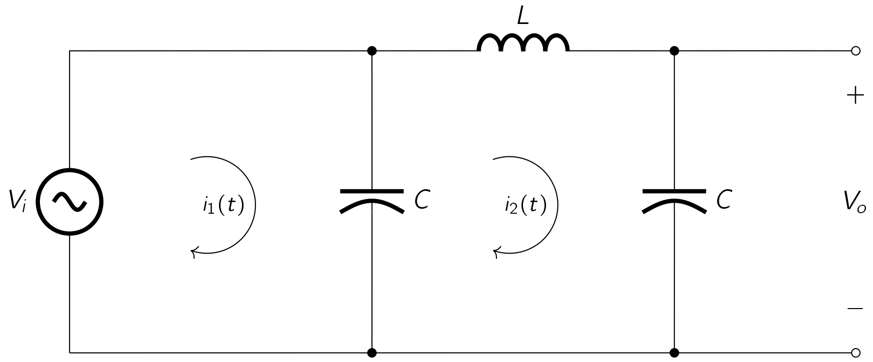

Figure 1: passive π filter

Solution

In this example, we find the transfers funcion by first transforming the circuit into the s domain

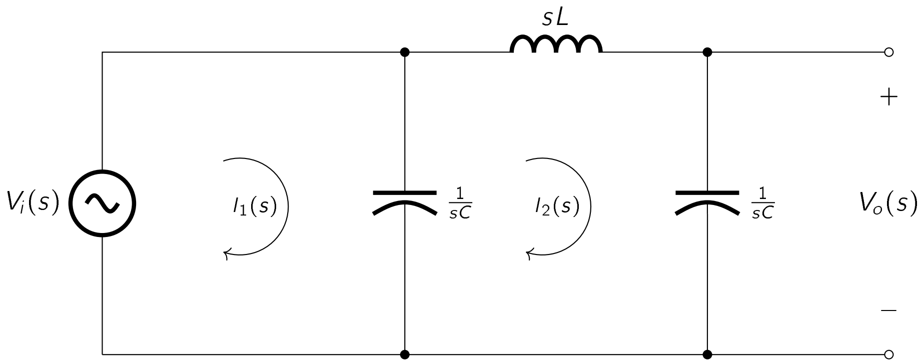

Figure 2: passive π filter

By KVL (Kirchhoff Voltage Law), the sum of voltage transforms around the loop is

$$V_i(s) - \frac{1}{sC}(I_1(s) - I_2(s)) = 0$$
$$V_i(s) - \frac{1}{sC}I_1(s) + \frac{1}{sC}I_2(s) = 0$$
$$\frac{1}{sC}I_1(s) - \frac{1}{sC}I_2(s) = V_i(s) \tag{1}$$

the same step for loop 2 gives

$$
\begin{aligned}
-\frac{1}{sC}[I_2(s) - I_1(s)] - sLI_2(s) - \frac{1}{sC}I_2(s) &= 0 \\
-\frac{1}{sC}I_2(s) + \frac{1}{sC}I_1(s) - sLI_2(s) - \frac{1}{sC}I_2(s) &= 0 \\
\frac{1}{sC}I_1(s) - \left[sL + \frac{1}{sC} + \frac{1}{sC}\right] I_2(s) &= 0 \\
\frac{1}{sC}I_1(s) - \left[sL + \frac{2}{sC}\right] I_2(s) &= 0 \qquad (2)
\end{aligned}
$$

From (1) and (2)

$$
v_O = \begin{cases} \frac{1}{sC}I_1(s) - \frac{1}{sC}I_2(s) = V_i(s) \\ \frac{1}{sC}I_1(s) - \left[sL + \frac{2}{sC}\right] I_2(s) = 0 \end{cases} \qquad (3)
$$

We have a 2x2 linear system, solving for $I_1(s)$ in 3(b) we get

$$
\begin{aligned}
\frac{1}{sC}I_1(s) &= \left[sL + \frac{2}{sC}\right] I_2(s) \\
I_1(s) &= sC\left[sL + \frac{2}{sC}\right] I_2(s) \\
I_1(s) &= (s^2CL + 2)I_2(s) \qquad (4)
\end{aligned}
$$

Replacing (4) in (3a) yields

$$
\begin{aligned}
\frac{1}{sC}\left[s^2CL + 2\right] I_2(s) - \frac{1}{sC}I_2(s) &= V_i(s) \\
\left[sL + \frac{2}{sC}\right] I_2(s) - \frac{1}{sC}I_2(s) &= V_i(s) \\
\left[\frac{s^2LC + 2}{sC}\right] I_2(s) - \frac{1}{sC}I_2(s) &= V_i(s) \\
\left[\frac{s^2LC + 2 - 1}{sC}\right] I_2(s) &= V_i(s) \\
I_2(s) &= \frac{sC}{1 + s^2LC}V_i(s) \qquad (5)
\end{aligned}
$$

Now, the output voltage is

$$
V_o(s) = \frac{1}{sC}I_2(s) \qquad (6)
$$

Replacing (5) in (6) yields

$$
V_o(s) = \frac{1}{sC}\left[\frac{sC}{1 + s^2LC}\right]V_i(s) = \left[\frac{1}{1 + s^2LC}\right] \qquad (7)
$$

The transfer funcion is

$$
H(s) = \frac{V_o(s)}{V_i(s)} = \frac{1}{1 + s^2LC} \qquad (8)
$$

The frequency response is obtained by substituting $s = j\omega$ into $H(s)$

$$
H(\omega) = H(s)\Big|_{s=j\omega} = \frac{1}{1 + (j\omega)^2LC} = \frac{1}{1 - \omega^2LC} \qquad (9)
$$

The magnitude response is given by the aboslute value of $H(\omega)$

$$A_v = |H(\omega)| = \left|\frac{1}{1-\omega^2 LC}\right| = \frac{1}{\sqrt{1+(\omega^2 LC)^2}} = \frac{1}{\sqrt{1+\left(\frac{\omega L}{\frac{1}{\omega C}}\right)^2}} = \frac{1}{\sqrt{1+\left(\frac{Z_L}{Z_C}\right)^2}} \tag{10}$$

Solving (10) for L

$$\begin{aligned}
A_v\sqrt{1+\left(\frac{\omega L}{\frac{1}{\omega C}}\right)^2} &= 1 \\
\sqrt{1+\left(\frac{\omega L}{\frac{1}{\omega C}}\right)^2} &= \frac{1}{A_v} \\
1+\left(\frac{\omega L}{\frac{1}{\omega C}}\right)^2 &= \left(\frac{1}{A_v}\right)^2 \\
\left(\frac{\omega L}{\frac{1}{\omega C}}\right)^2 &= \left(\frac{1}{A_v}\right)^2 - 1 \\
\frac{\omega L}{\frac{1}{\omega C}} &= \sqrt{\left(\frac{1}{A_v}\right)^2 - 1} \\
\omega L &= \frac{1}{\omega C}\sqrt{\left(\frac{1}{A_v}\right)^2 - 1}
\end{aligned} \tag{11}$$

So

$$L = \frac{1}{\omega^2 C}\sqrt{\left(\frac{1}{A_v}\right)^2 - 1} \tag{12}$$

For $f = 100\,\text{Hz}$, $A_v = 0.1$ and $C = 100\,\mu\text{F}$

$$L = \frac{1}{2\pi \cdot (100)^2 \cdot (100 \times 10^{-6})}\sqrt{\left(\frac{1}{0.1}\right)^2 - 1} \approx 2.52\,\text{H} \tag{13}$$

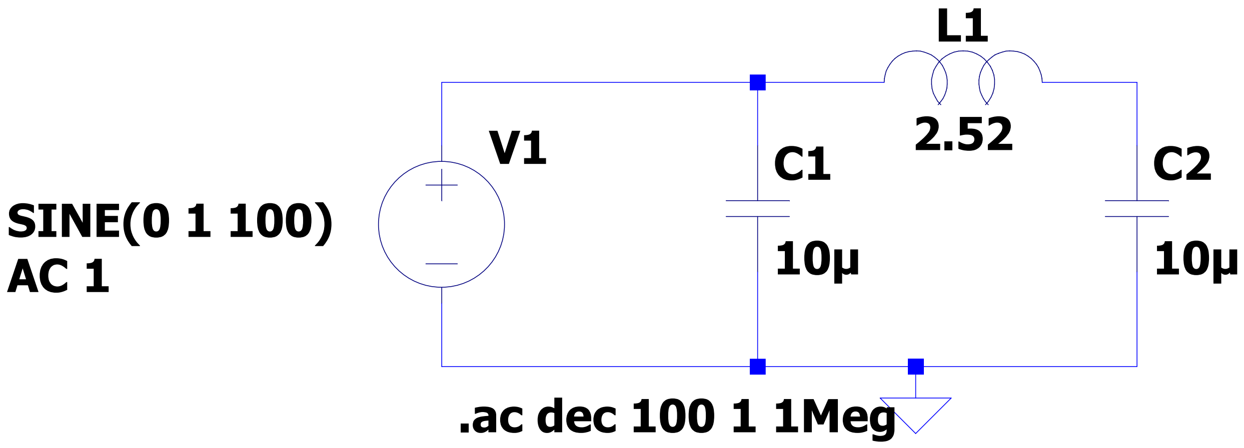

Figure 3:

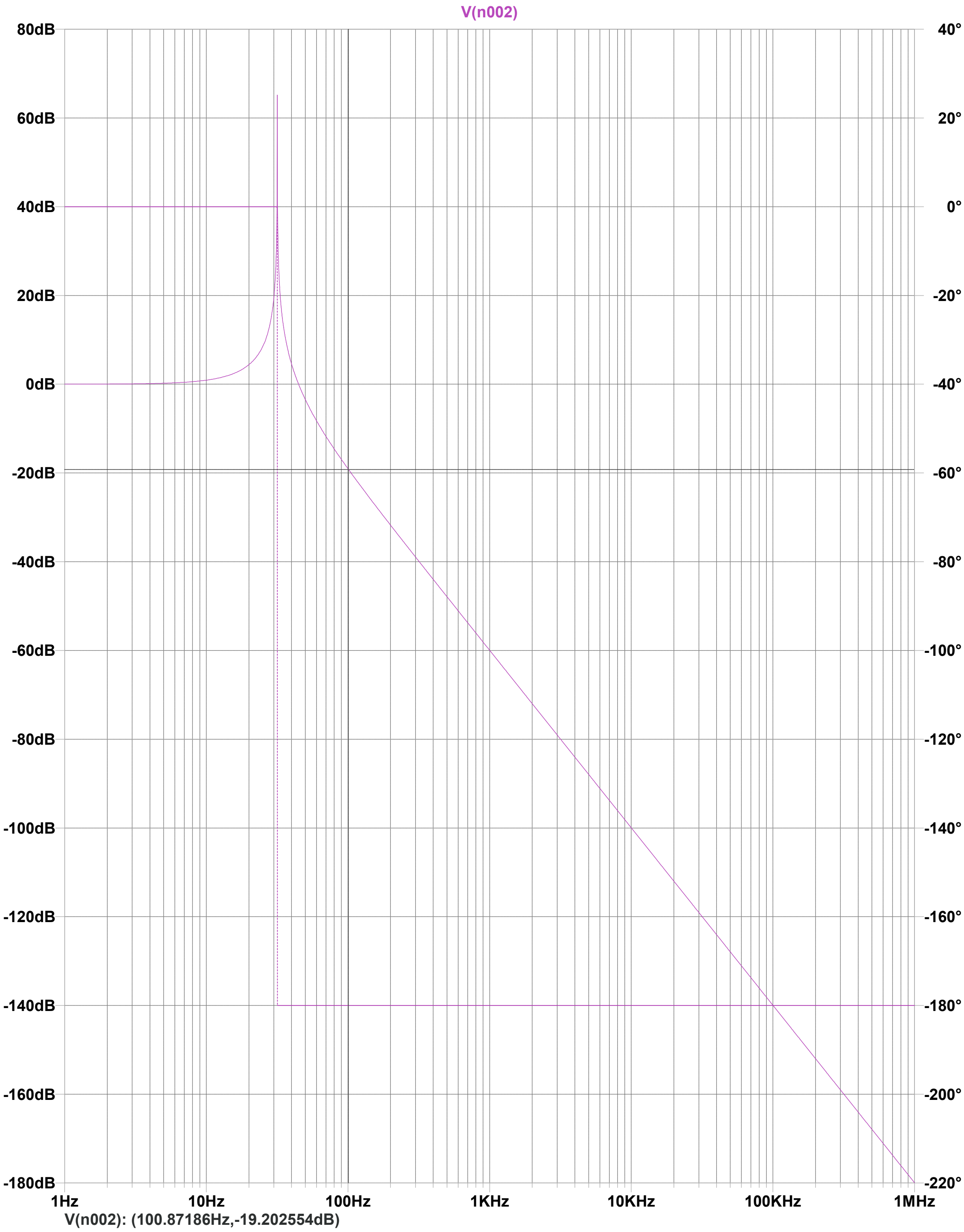

Figure 4:

A we can see in the graph on bottom left side, the V(n002) probe shows that at 100 Hz the gain in dB is ($-19.2 \approx -20dB$)
Remember that the Gain in dB is given by

$$G_{dB} = 20\log_{10} A_v \tag{14}$$

Solving for A_v

$$A_v = 10^{\frac{G_{db}}{20}} = 10^{\frac{-20dB}{20}} = 0.1 \tag{15}$$